

M.A.Seenivasan

Researcher and Teacher (Status*: Ph.D Thesis submitted)

Department of Electronics and Communication Engg.

National Institute of Technology Meghalaya

Sohra, Meghalaya, INDIA-793108.

Teaching - ECE and Electronics & Computing Engineering Courses.

Research - Future of VLSI and Computing: Neuromorphic Computing, From CMOS scaling to AI-Driven Spiking Neural Networks (SNNs).

Tools Proficiency - Hardware: Cadence EDA Tool, Vivado, FPGA/ASIC

Neural Computing Simulations Software Tools: Python, Matlab, C.

Publications - SCI-Journals: First Author 5+ and Co-author 1; Confs: 2+



COMMUNICATION DETAILS



: +91-9962224603(M)



: maseenivasan@gmail.com

WEBPAGE : [M.A.Seenivasan](#)

R-GATE : [23.4 R-Interest Score, 16 Citations, 2 h-index](#)

LINKEDIN : [seenivasan@iiitp.ac.in](#)

G-SCHOLAR : [Citations: 16, h-Index: 2, i10-Index:0](#)

ORCID-ID : [0000-0002-6104-2250](#)

WEBOS-ID : [ABF-9111-2022](#)

SCOPUS-ID : [56340679300](#)

RESEARCH AREAS AND TEACHING COURSES

- **Neuromorphic Modeling, Computing, Circuits and Systems**- Theoretical & Computational Neuroscience, Neuron and Synapse Modeling, CMOS Implementation of Spiking Neural Network (SNN) to the development of Neuromorphic AI-Driven Chip Engineering.
- **Electronics (Analog and Digital) Fundamentals**- Device Physics, Analog and digital building blocks, Nano-Scale CMOS Circuit Design and Implementation, and VLSI Signal processing.
- **Computer Engineering**- Introduction to Programming (using C or Python), Digital Logic & Microprocessors, Computer Architecture, Networks, IoT & Embedded Systems.

EDUCATION

Degree	University	Year	CGPA
Ph.D.	MICROELECTRONICS & VLSI RESEARCH GROUP, Department of ECE, National Institute of Technology Meghalaya , India (NIRF:83)	2026* (TS)	9.67
M. E (Master of Engineering)	APPLIED ELECTRONICS, College of Engineering Guindy, Anna University Main Campus , Chennai, India (NIRF:20)	2014	8.07
B.E. (Bachelor of Engineering)	ELECTRONICS AND COMMUNICATION ENGINEERING, Anna University , Chennai, India	2009	76.0%
H.S.S. (12 th)	MATHEMATICS AND SCIENCE, Tamil Nadu State Board , Thiruvallur District, Tamil Nadu, India (secured 100% in Biology)	2005	91.5%
High sch. (10 th)	ALL SUBJECTS, Tamil Nadu State Board , Tamil Nadu, India	2002	93.2%

WORK EXPERIENCE

Employer	Post Held	Duration	
		From	To
NIT Meghalaya	Research Scholar	28-07-2021	08-06-2026
NIT Meghalaya	SRF	01-01-2024	19-05-2026
NIT Meghalaya	JRF	28-07-2021	31-12-2023
SRMVEC Chennai(NIRF:151-200)	Asst.Prof (Ordinary scale)	01-06-2015	27-07-2021

RESEARCH

Interest

Neuro theory, Physics of Learning, **Temporal Dynamics and Efficient Learning in Spiking Neural Networks (SNNs)**, **CMOS-based hardware implementation of SNNs**, Mixed-signal circuits, Memristor-based design; **Bridging the Gap between Neuromorphic Computing and Machine Learning**, and architectures towards Neuromorphic AI-Driven chip engineering.

Vision and Passion

The present work covers the hardware implementation of a spiking neural network inspired by neuroscience principles. My long-term research objective is to develop scalable neuromorphic hardware architectures for edge AI and intelligent embedded systems. Specific aspects I am looking to explore include **algorithm-to-hardware mapping (e.g., SNNs)**, **Circuit-level design (e.g., analog neurons, memristors)**, and **Applications (e.g., edge AI, robotics, or brain-computer interfaces)** that provide specific **topologies and design considerations to tailor the project**.

These systems will target next-generation electronics and computer engineering applications, including **intelligent Internet of Things (IoT) platforms and embedded systems**, enabling low-power edge intelligence and real-time adaptive computation.

Publication in Journals

1. M.A. Seenivasan, Prabir Saha, Shubhankar Majumdar, and Gautam Srivastava. Biologically inspired stdp hardware learning architecture for neuromorphic perception computing. *Computers and Electrical Engineering Journal, Elsevier*, 2026 (**SCI Q1- Revision submitted.**)
2. M.A. Seenivasan, Prabir Saha, Shubhankar Majumdar, and Alak Majumdar. A compact mathematical model and hardware implementation of spike-frequency adaptation for low-power neuromorphic computing. *IEEE Transactions on Nano Technology (TNANO)*, IEEE, 2026 (**SCI - Published.**)
3. M.A. Seenivasan, R.U. Ahmed, Prabir Saha, and Shubhankar Majumdar. Synaptic plasticity dynamics of a biological neuron from a model of excitatory and inhibitory neurotransmitter releases towards hardware implementation of stdp in cmos. *Neurocomputing, Elsevier*, 2026 (**SCI Q1-Published.**)
4. M.A. Seenivasan, Prabir Saha, Shubhankar Majumdar, and Gautam Srivastava. Design of synaptic interconnection of a sensory lif neuron in cmos for high-speed in-sensor neuromorphic systems. *IEEE Sensor Journal (2026)*, IEEE, 2026 (**SCI-Q1)-Published.**
5. M.A. Seenivasan, Prabir Saha, and Shubhankar Majumdar. Mathematical model and its realization of subthreshold internal membrane dynamics and implementation in cmos neuromorphic systems. *The International Journal of Numerical Modeling: Electronic Networks, Devices and Fields (IJNM)*, Wiley, 2026 (**SCI-Q3/Q2)-Published.**
6. M.A. Seenivasan, A.V. Parekkattil, R.U. Ahmed, and P. Saha et al. Biologically inspired tonic and bursting lif neuron model for spiking neural network: a cmos implementation. *Microsyst Technol* 31, 475–489 (2025), *Springer-Nature*, 2025 (**SCI-Q2/Q3)-Published.**
7. R. U. Ahmed, H. R. Thakur, M.A. Seenivasan, and P. Saha. Power-efficient vlsi realization of decimal convolution algorithms for resource-constrained environments: A design perspective in cmos and double-gate cmos technology. *Microsystem Technologies, Springer-Nature*, 2024 (**SCI-Q2/Q3)-Published.**

Publication in Conferences

1. M. A. Seenivasan, Geetima Kachari, Shubhankar Majumdar, and Prabir Saha. Low-voltage differential signaling (lvds) modeling for synapse-to-neuron neuromorphic interconnects. In *30th International Symposium on VLSI Design and Test*, 2026. (**VDAT-Flagship conference**) [Submitted]
2. M. A. Seenivasan and P. Saha. The energy-efficient and distortion blocking leaky integrate-and-fire neuron model. In *IEEE Int. Conf. on Energy, Power and Environment (ICEPE)*. IEEE, 2024, (**IEEE Conference**)

-
3. P. Saidesh Kumar and M. A. Seenivasan. A 32×32 cmos image sensor: Tested using process and temperature compensated voltage controlled current source. In *18th International Symposium on VLSI Design and Test*, pages 1–6, 2014, (**VDAT-Flagship conference**)
 4. P. Saidesh Kumar and M. A. Seenivasan. Wide dynamic range cmos image sensor for star trackers. In (**Cadence Design Contest 2014**), at *Cadence Design Systems Bangalore, India*. Cadence open project category, 2014

Publication in Book Chapters

1. M.A. Seenivasan and Prabir Saha. (writing) a book chapter on the introduction of neuromorphic hardware design for neuromorphic edge ai computing. In *Neuromorphic Computing and System Design for Next-generation VLSI*, 2027

Research Group

1. **Prof. (Dr.) Gautam Srivastava**, is with Brandon University, Canada, Sogang University, Republic of Korea, China Medical University, Taiwan, and Chitkara University, Punjab, India.
2. **Dr. Prabir Saha**, Associate Professor, is working at NIT Meghalaya, India.
3. **Dr. Shubhankar Majumdar**, Assistant Professor, is working at NIT Meghalaya, India.
4. **Dr. Alak Majumdar**, Assistant Professor, is working at NIT Arunachal Pradesh, India.
5. **Mr. Adarsh V. Parikattil** is pursuing Ph.D at Heidelberg University, Germany.

Project Proposals

Neuromorphic computing is a frontier area in deep tech, and India is actively increasing funding in this domain, especially in AI hardware and next-generation computing. I am **planning a proposal on developing low-power AI hardware** (e.g., SNN CMOS/VLSI implementation) and edge computing applications in the following funding agencies

1. AICTE - Schemes: MODROBS, RPS, ATAL
2. UGC - Minor/Major Research Projects; ANRF under PMECRG - Core Research Grant (CRG)
3. DST - Schemes: INSPIRE, ICPS, PURSE; MeitY - Focus: AI hardware, neuromorphic computing
4. State-level funding - TNSCST, MSME, etc, INDUSTRY FUNDING

Scholarships Received

1. Receiving **PhD scholarship/assistantship** by the Ministry of Human Resource Development (MHRD), Government of India, at NIT Meghalaya.
2. Received **Best Project award** by SRM Funding Agency, Department of ECE, SRM Chennai, from 2019 to 2020.

References

1. **Dr. Prabir saha**, Associate Professor, Department of ECE, NIT Meghalaya, India
E-mail: prabir.saha@nitm.ac.in, Ph: (+91) 9485177005.
2. **Dr. Shubhankar Majumdar**, Assistant Professor, Department of ECE, NIT Meghalaya, India
E-mail: shub@nitm.ac.in, Ph: (+91) 7550197054
3. **Dr. Gautam Srivastava**, Professor, Department of CSE and Mathematics, Brandon University, Canada E-mail: srivastavag@brandonu.ca, Ph: +1 (204) 727-9748

TEACHING AND DEVELOPMENT TRAINING AND OUTREACH PROGRAM AT SRMVEC DURING 01-06-2015 TO 30-06-2021

AICTE Staff ID:1-2680021928

Sl. No.	Outreach Program	Duration	
		From	To
1.	NITTT Technology-Enabled Learning & Lifelong Self Learning	Jul-2020	Oct 2020
2.	NITTT Orientation Towards Technical Education and Curriculum Aspects	Jul-2020	Oct 2020
3.	NITTT Professional Ethics & Sustainability	Feb-2020	May 2020
4.	NITTT Communication Skills, Modes & Knowledge Dissemination	Feb-2020	May-2020
5.	NPTEL Microelectronics: Devices to Circuits	July 2019	Oct 2019
6.	NPTEL Outcome-based pedagogic principles for effective teaching	August-2018	Oct 2018
7.	PMMMNMST sponsored One Month Faculty Induction Training Programme	May-2018	May 2018
8.	NPTEL VLSI Physical Design	Jan-2017	Apr 2017

WORK RESPONSIBILITIES AT SRMVEC DURING 01-06-2015 TO 30-06-2021

ADMINISTRATIVE RESPONSIBILITIES

Sl. No.	Responsibility	Duration	
		From	To
1.	Incharge Newsletter and report preparation	August-2016	June 2021
2.	Incharge Fine-Arts-Club and social media	August-2017	June 2018
3.	Warden SRMVEC Boys Hostel 1	July-2016	June 2017
4.	Faculty Advisor B.E SRMVEC (Alternate semester)	July-2016	May-2021
5.	Member Exam Cell Incharge	01-06-2020	30-06-2021
6.	Member Departmental Research Committee	August-2019	May 2020

FDP/EVENTS ORGANIZED

Sl. No.	Title	Funding Agency	Role	Duration	
				From	To
1.	End Semester Practical Examination conduction	SRM	Incharge	01-06-2020	30-06-2021
2.	A webinar on 'Bio Sensors and its Applications'	SRM	Incharge	19-11-2020	19-11-2020
3.	National Level Technical Symposium TARAS 2K20	SRM	Incharge	15-02-2020	15-02-2020
4.	A Seminar on 'Opportunities in Civil Services for Engineers'	SRM	Coordinator	18-07-2018	18-07-2018
5.	A Faculty Development Programme on 'Electronic Circuit-I'	SRM	Coordinator	08-06-2017	09-06-2017
6.	An Expert lecture on 'Next Generation Networks'.	SRM	Incharge	06-02-2017	06-02-2017
7.	Value Added Course on 'Embedded and Real Time Systems'	SRM	Coordinator	30-01-2017	01-03-2017

Regards

M.A.Seenivasan